

Safety inputs: 2 e-stops + 2 lidars, 24 V through PC817 – shield for Arduino UNO R4 WiFi / UNO Q

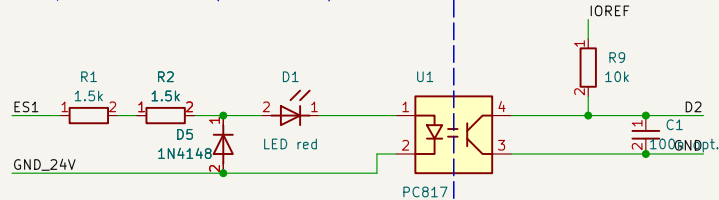
E-stop loops: $2 \times 1.5 \text{ k} \rightarrow 6.9 \text{ mA}$ (109 mW per resistor at 28.8 V). Lidar inputs: $2 \times 2.2 \text{ k} \rightarrow 4.7 \text{ mA}$ (75 mW). All resistors 1/4 W.

No current (button pressed, zone occupied, wire cut, 24 V lost) $\rightarrow$ pin HIGH = STOP. F1 protects the 24 V wires going out to the buttons.

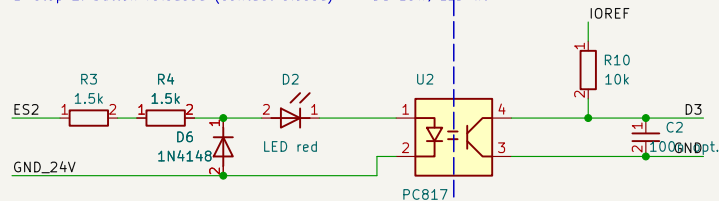
GND_24V and GND are NOT connected: this is the isolation barrier. Pull-ups go to IOREF (5 V on UNO R4 WiFi, 3.3 V on UNO Q), never to 5V.

C1–C4 are OPTIONAL: fitted, an opening shorter than about 1.5 ms is not seen by the Arduino, and every stop is seen about 1.6 ms later.

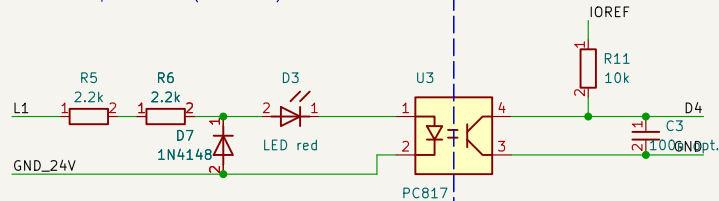
E-stop 1: button released (contact closed) $\rightarrow$ D2 LOW, LED lit



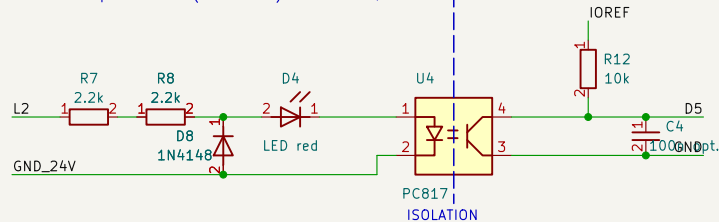
E-stop 2: button released (contact closed) $\rightarrow$ D3 LOW, LED lit



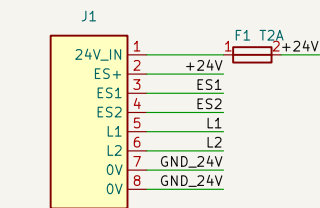
Lidar 1: output at 24 V (zone clear) $\rightarrow$ D4 LOW, LED lit



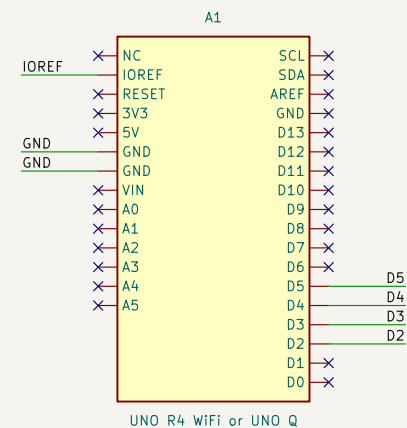
Lidar 2: output at 24 V (zone clear) $\rightarrow$ D5 LOW, LED lit



ISOLATION



24V, e-stops, lidars



PROTO-SHIELD VARIANT: 2.54 mm grid footprints, 8-way terminal
Generated by generate.py – edit the script, not this file

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